

Time Sensitive Networking Learning Report

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Abstract—Time-Sensitive Networking (TSN) comprises a set of IEEE 802.1 standards designed to enable deterministic communication over Ethernet, providing bounded latency, low jitter, and high reliability. TSN achieves these guarantees through mechanisms including time-aware scheduling, traffic shaping, and precise time synchronization. This report investigates the extension of TSN into wireless domains by discussing synchronization mechanisms in IEEE 802.11 (Wi-Fi) and 5G networks. Two representative studies are reviewed and compared to evaluate their approaches and performance in TSN-enabled wireless scenarios. The results show that, in a TSN Wi-Fi systems, shorter beacon intervals improve synchronization accuracy among devices. In contrast, the 5G TSN study demonstrates that synchronization performance is highly sensitive to system parameters, where small variations can lead to significant error deviations. Furthermore, Sionna is identified as a suitable open-source framework for supporting TSN research through link-level simulation, particularly for analyzing the impact of wireless channel conditions on deterministic communication. A case study on physical-layer abstraction is presented to illustrate an efficient computational approach for evaluating channel behavior without requiring physical layer processing.

Index Terms—TSN, Wireless, IEEE 802.11, 5G Cellular, Sionna, Network Simulation

I. INTRODUCTIONS

The evolution of networking technologies is creating rapid demands for deterministic communication, which is capable of low latency, high reliability, and efficient resource management for data transmission [1]. In particular, industry fields have been a potential application for this level of performance due to their heterogeneous nature from automation or remote control in the manufacturing process, moreover, industry 5.0 [2] is apparent with the extensive implementation of the Internet of Things (IoT) [2] and the 5G cellular network [3]. We focus on one specific standard to achieve this level of communication in this learning report, namely Time Sensitive Networking [1] (TSN). The following sections introduce the fundamental concepts of TSN and two wireless applications, including IEEE 802.11 [3] and 5G cellular [3].

A. Time Sensitive Networking

Time-Sensitive Networking (TSN) is a series of standards based on IEEE802.1 [3], the objective is to enable deterministic communication within Ethernet networks, we explore TSN in wireless scenarios in the following chapters. TSN merges critical and best-effort network traffic in a unified network architecture to retain complexity, even with various traffic classes. The general techniques for TSN in Ethernet applications are time-aware scheduling, traffic shaping, and clock synchronization to maintain bounded latency, low jitter,

and high reliability [3]. This learning report introduces the general structure of TSN with the core components, and we primarily discuss two wireless topologies under the TSN infrastructure including Wi-Fi and 5G applications.

TSN is a critical enabler for deterministic networking in industrial applications including automation and robotic systems [2] due to the manufacture process requires stringent end-to-end latency to improve efficiency and enhance reliability. Therefore, the following chapters in this learning report discuss and analyze TSN in wireless implementations from the perspective of industrial users.

B. IEEE 802.11

IEEE 802.11 is the essential standard of Wi-Fi that has been evolving since its initial release including IEEE 802.11 with ac (Wi-Fi 5), ax (Wi-Fi 6/6E), and current generation IEEE 802.11be (Wi-Fi 7). The new release for each generation with the improvement in throughput, frequency band allocation, multiple user support, and orthogonal methodologies [3]. TSN can also coordinate with IEEE 802.11 considering it supports synchronized communication from IEEE 802.11ax (Wi-Fi 6), and potential bounded latency capability in IEEE 802.11be (Wi-Fi 7) [1], we have a comprehensive discussions in Section II about TSN application in Wi-Fi scenario and a case study.

The integration of TSN and IEEE 802.11 provides new opportunities for enabling real-time wireless communication, where the industrial environment can utilize the sensors and actuators with low latency by clock synchronization techniques. Moreover, TSN with IEEE 802.11 can utilize time-aware scheduling [3] to distribute the packets in a precise time, this technique has the original definition in standard IEEE802.1Qbv based on the wired network, however, this learning report concentrates on wireless implementation combined with TSN.

C. 5G Cellular

5G represents the fifth generation of cellular network standardized by 3GPP [4], it contains more advanced characteristics than 4G including ultra-reliable low-latency communication (URLLC), enhanced mobile broadband (eMBB), and massive machine-type communication (mMTC), we discuss TSN in 5G application with its corresponding case study in Section III. Moreover, IEEE 802.1 Ethernet network architecture is the foundation of the TSN standardization that differentiates from the cellular network, therefore, it requires the translator to ensure compatibility between 5G and TSN, we explain

the system architecture and of the TSN deployment in a 5G network in Section III.

The combination of TSN and 5G can benefit the industrial users to enable industry 5.0 [2] by employing in collaborative robots, automation, and vehicle-to-everything (V2X) communications [1], which can ensure the efficiency and time precision in data transfer among the devices. TSN with 5G can also impact media fields, especially in video/audio streaming, the low latency with high reliability creates a real-time data transmission to allow better experiences for consumers to watch live events across all the platforms.

D. Sionna

Sionna is an open-source library that provides link-level simulation capabilities powered by Python and TensorFlow [5]. It primarily supports experimentation in communication systems and features GPU acceleration, enabling the integration of Machine Learning (ML) techniques within the physical layer to optimize performance across various scenarios. Due to the complexity of physical-layer processes and hardware constraints, this learning report does not explore these aspects in depth. Instead, Section IV focuses on a tutorial covering Physical Layer Abstraction, emphasizing wireless channel transmission and its evaluation.

Sionna is a highly capable framework that offers modular components for constructing sophisticated communication models across diverse applications. It can simulate signal fading, path loss, and other forms of interference [6], thereby providing a realistic simulation environment. Furthermore, Sionna incorporates several standardized channel models, including Additive White Gaussian Noise (AWGN), 3GPP 38.901, and 5G New Radio [7]. The library also supports multi-antenna configurations, enabling the evaluation of transmission reliability. Sionna consists of three primary modules: Sionna RT (Ray Tracing), Sionna PHY (Physical), and Sionna SYS (System) [5]. The SYS and PHY modules are developed using TensorFlow, while the RT module is built upon Mitsuba 3 and Dr.Jit. This report does not elaborate on the underlying mechanisms of these modules, but instead provides a high-level overview based on the PHY abstraction tutorial.

Several research efforts have utilized Sionna to simulate wireless communication systems, demonstrating its potential for future studies. A study on a 5G New Radio multi-user MIMO OFDM system [7], further evaluated in [6], compared multiple 3GPP 38.901 channel models (Urban Micro, Urban Macro, Rural Micro, and Rural Macro) using Block Error Rate (BLER) as the performance metric. The results indicate that longer code lengths exhibit improved error tolerance. In another study, researchers implemented two high-level modulation schemes under trainable and non-trainable configurations in a 5G New Radio system, demonstrating that higher bits per symbol can achieve better performance conditionally.

To initiate the research on TSN, this learning report leverages Sionna to simulate a wireless channel conditions with PHY abstraction. Although this implementation alone cannot capture end-to-end latency or evaluate a complex TSN system,

the wireless channel can be one of the prerequisites to realize a lower bound for latency and jitter. In a word, the implementation of Sionna represents the first solid foundation before constructing a complete TSN system in Sionna.

E. Sionna and TSN

We select Sionna as the implementation platform for our TSN research due to its flexibility in constructing modular simulations across the physical and link layers. While TSN mechanisms primarily operate at Layer 2 (Data Link Layer) [3], their performance in wireless environments including Wi-Fi and 5G is strongly influenced by physical-layer behavior.

In the case studies analyzed in this report, TSN is integrated with Wi-Fi and 5G systems, where the evaluation of synchronization performance inherently depends on underlying channel conditions. Channel characteristics, including interference and propagation effects, play a critical role in determining whether TSN requirements can be satisfied in wireless deployments, and Sionna is capable to simulate these conditions to determine the performance of a wireless network.

Therefore, Section V presents an initial implementation using Sionna to investigate the feasibility of identifying suitable channel conditions before we can merge a TSN wireless into the network model. By leveraging Sionna's physical-layer modeling capabilities, this work aims to provide a more realistic assessment of wireless channel behavior and its impact on TSN performance. Although the current evaluation does not yet incorporate end-to-end latency analysis, it establishes a foundational step toward building and evaluating a more comprehensive TSN framework within a realistic wireless simulation environment. The use of PHY abstraction in this study is primarily motivated by computational resource constraints; nevertheless, the obtained results remain indicative and provide a useful reference for future work.

F. Objective of Learning Report

This learning report presents two sections of the study process. The theoretical part covers the implementation of Time-Sensitive Networking (TSN) in Wi-Fi and 5G applications, comparing these two scenarios in terms of scalability and complexity, as well as the aspects that can and cannot be concluded yet. The practical part describes a Sionna simulation used to build a physical layer (PHY) abstraction model based on an available tutorial. This simulation does not directly evaluate TSN in wireless networks but serves as a foundation for subsequent doctoral research.

Furthermore, this report is intended for instructional purposes for students who wish to study TSN or use Sionna to simulate their own projects. Section IV provides brief instructions for installing and troubleshooting the virtual environment required to launch Sionna. The objective of this report is not only to address critical topics related to TSN in wireless networks and Sionna simulation but also to document the process of learning and conducting a research topic.

G. Learning Report Structure

This learning report has the following structure: Section I introduces the general definitions of TSN, IEEE 802.11, and 5G cellular. Section II is the discussion in IEEE 802.11 with TSN integration. Similarly, Section III presents the implementation of 5G cellular with TSN, and a comprehensive comparison between two case studies is presented in Section IV. Section V introduces Sionna and PHY abstraction as the practical part of the learning process. Section VI concludes this learning report and presents some future works based on TSN in wireless networks and Reliable and Available Wireless (RAW).

II. TSN FOR WI-FI APPLICATIONS

This chapter introduces the first wireless field that TSN can contribute to, there are three sections including latency classification under industrial wireless communication. The system architecture contains all the elements composed in a TSN Wi-Fi implementation environment, finally, the case study discusses an evaluation of TSN within Wi-Fi setup based on references [8] [9], which summarize their measurement setting, performance indicators and results.

A. Latency Classifications

The communication in networking always has latency within data transmission, we only discuss the range of latency related to meeting TSN requirements in industrial applications. Table 1 shows the various circumstances based on the range of latency, throughput, and power supply, we select the partial elements from the essay [9] because we intend to concentrate on the particular essential metrics related to TSN implementation in IEEE 802.11. We explain each scenario and the technical specification for its throughput and latency range, power supply is an additional attribute that we consider it be useful but we do not cover this topic in this learning report.

1) *Robotic Sensors and Remote Control*: These robotic sensors and remote controls are in an industrial Internet of Things (IoT) environment, where they are typically part of the automation system. Car manufacturers utilize robotic arms to operate painting or some precise workflow when they produce the vehicles, the internal sensors must share the status in real-time with to control center to monitor the entire production line with correct configuration and operations. Therefore, they should have stringent latency as lowest to 0.1 ms for critical wireless communication [9]. Some periodical updates might have more tolerance on higher latency up to 100 ms, the important thing is to maintain the system with consistent precision operations with significantly low latency.

The throughput is the transferred bits divided into duration, the industrial IoT can have various data to transfer, and it can have low throughput if it is just configuration information including certain parameters to produce the car in a manufacturing environment. However, the throughput can also become high if the video or high-resolution images are the payload in data, it depends on which robotic sensors we want to enable deterministic communication. Finally, most manufacturers are operating the production with main power as the power grid

is widely available for industrial users, however, we do not measure power consumption in this learning report other than an informative element.

2) *Mixed Reality and Holographic Contents*: The mixed reality and holographic contents are promising for media content industries, which consume high computational power, it needs to transfer the high data volume even for a TSN environment due to the primary payload is the video. Additionally, mixed reality has motion-capture for some interactive experiences, that require strict delay for usable operations [9]. The lowest latency range does not reach the same as the robotic sensor and remote control do due to this scenario frequently applies to consumers, but it should maintain within 10 ms to ensure functional experiences for mixed reality and holographic content users.

Naturally, the large data volume leads to high throughput, and it should be a stable stream since video data constantly generates huge data traffic. The source of power is the battery since we consider mixed reality and holographic contents in the mobile hardware, however, we do not consider the optimization of power consumption within the TSN environment, the video streaming in devices can drain the battery significantly due to the intensive computational processing.

3) *Professional Media and Live Streaming*: The media streaming in the professional field requires accurate feedback from the recording instruments, vocalists, and some electronic instrumentals, the speakers in the control must align and collaborate because any time difference or high latency can lead to defective mixing, more than 5 ms can cause the complete mismatch between vocal and instrumental [9]. Therefore, the communication in audio and live streaming must have a strict latency threshold and it should maintain as low as possible consistently.

The throughput in this scenario has a range of low to high since the audio recording or streaming does not generate a large volume of data and traffic. However, some cases combine the video and audio to create the large data volume including living streaming in the venues, thus, the throughput can be various, the more important thing is the latency range should retain 1 ms to avoid inconsistent production results. As for the power supply, some professional equipment always connects to a power grid, and some live streaming devices from the user side can be mobile phones or tablets with batteries as their power source.

4) *Health Monitoring*: The health monitoring has become an essential subject for cutting-edge technologies, and TSN is not an exception here, some medical-graded devices have multiple sensors to measure and transfer the physical status of patients to doctors or control centers [9]. The connect health monitoring system depends on deterministic wireless connectivity and it can even save patients' lives in critical condition, although it does not have the lowest latency range in Table 1, the faster that critical healthy data can reach doctors during an emergency, the more possibility they can save lives with help of deterministic communications.

We consider the primary data transmission in health monitoring scenarios to be relatively low including heart rate, blood pressure, or oxygen level from the patients, most of these indicators are number values but audio or video contents. However, the reliability of this scenario should be highest to save people's lives with reliable links, some medical devices are battery-powered and they allow patients to mobile freely, ICU rooms can have more types of equipment with a power grid as the source, we primarily consider mobile devices as the application in this scenario.

5) *Multiplayer Gaming Experiences*: The gaming industry has been developing deterministic networking features especially modern online video games are demanding lower latency and more capable networks. Low latency is not just an important factor in playing online games, but also a precise guarantee for professional players, we intend to discuss the latency between the various online players when they are using independent PC other than the delay between the peripherals and games [9].

The throughput of the gaming communication is high due to the footage can be equivalent to video recordings, and it requires strict and low latency for reliable and precise control for players. The latency range is located between 20 and 100 ms depending on the type of online game, typically, First-person-shooting (FPS) games require the lowest latency among players, and cooperation games might have more tolerance but still require reliable and low bounded latency, the frame rate is also a decisive aspect for throughput. We define the PC running the games as a desktop with an external power supply from the grid.

B. System Architecture

The system architecture of TSN can adapt the realistic implementations depending on the networking infrastructures, we visualize the general structure of a TSN in a wireless network in Figure 1 and simplify the original work [9] that has a hybrid model including wired and wireless components into a TSN and Wi-Fi network. We introduce all the components based on Figure 1 covering CUC, CNC, TSN-CS, TSN bridge, TSN access point, and wireless end stations.

1) *CUC*: Centralized User Configuration is responsible for collecting the configurations from wireless end stations in the application layer in a TSN Wi-Fi network, that contains the stringent latency range, traffic priority, and related Wi-Fi parameters [1]. It needs to forward this information to a centralized network configuration unit, the realistic implementation frequently utilizes edge computing devices to collect the configuration from industrial users (Wireless End Stations) to mitigate the latency.

2) *CNC and TSN-CS*: Centralized Network Configuration is the core component in a TSN Wi-Fi network, it receives the requirements and wireless end station configurations from CUC and schedules the transmission to meet the requirements of the TSN Wi-Fi network depending on the policies from wireless end stations. In terms of configuration subsystems

(TSN-CS), we depict two TSN-CS blocks to manage two Wi-Fi subnets, TSN-CS is responsible for managing the TSN-enabled access points with their TSN bridges; in other words, CNC decides the transmission schedules based on the requirements of CUC, and TSN-CS executes the decisions from CNC to TSN-enabled access points via TSN bridges [9], and TSN-CS must consider the capacity and resources from each access points attached to its TSN bridge while it is passing the configuration to CNC and CUC.

3) *TSN Bridge*: TSN bridge frequently coordinates with a Wi-Fi access point, it is responsible for enforcing deterministic operations including time-scheduling traffic, clock synchronization, and redundancy mechanism. TSN bridges also manage the upstream to TSN-CS and CNC, and downstream to access points and wireless end stations, the latest Wi-Fi 7 (IEEE 802.11 be) can even utilize a multi-user version of OFDM (Orthogonal Frequency-Division Multiplexing) with TSN bridge from MAC layer [1] to allow wireless end stations to have guaranteed latency and high reliability.

4) *TSN Access Point*: TSN access point is the practical entry for a TSN Wi-Fi network from wireless end stations, and it does not function as an ordinary access point when it features TSN capability because it must have time-aware scheduling transmission parameters including the gate for controlling the networking traffic flow depending on the priority. However, we do not discuss the mechanism of these TSN functions in this learning report other than illustrating the overall structure of a TSN Wi-Fi network and explaining the general functions of each element.

5) *Wireless end stations*: Wireless end stations are the end devices acting as listeners or talkers in a time-sensitive traffic flow, they can be robotic sensors, mixed reality devices, professional audio equipment, medical devices, and gaming PCs as the previous section introduced. They need to provide the TSN end station configuration to CUC by utilizing TSN access points, this is also where the scheduled transmission packets aggregate to realistic implementations. We discuss a case study in the next section where the PCs in the labs are the wireless end stations and the end-to-end latency in a TSN Wi-Fi network represents the time when a packet departs from the talker wireless end stations to the listener one.

C. Case-Study

This section contains the measurement setup including the configuration of the hardware environment and protocol running on the software as the first section depicted in Figure 2, the second section discusses the methodologies that the work [10] generates the data transmission and index to evaluate the performance, and the last two sections summarizes the results of their experiment and limitations.

1) Measurement Setup:

a) *PTP Synchronization*: This component intends to validate the accuracy of beacon-based synchronization based on the reference [10], a Precision Time Protocol (PTP) Linux implementation provides the time offsets of the wireless devices to ensure the effective synchronization. The wired device

Implementations	Throughput	Latency Range	Power Supply
Robotic Sensors and Remote Control	Low to High	0.1 to 100 ms	Power Grid or Battery
Mixed Reality and Holographic Contents	High	2 to 10 ms	Battery
Professional Media and Live Streaming	Low to High	1 ms	Power Grid or Battery
Health Monitoring	Low	20 to 100 ms	Battery
Multiplayer Gaming Experiences	High	2 to 10 ms	Power Grid

TABLE I: **Latency Classifications** *Implementations are the feasible scenarios that certain classes can utilize [9], throughput is the data rate for communication for TSN applications, Latency range is the bounded latency that each implementation should maintain, and power supply is the primary source for powering the wireless TSN.*

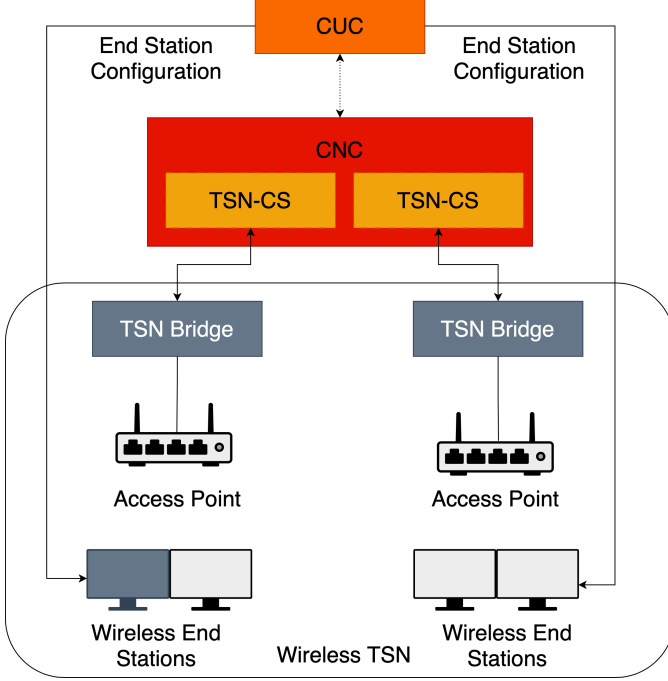


Fig. 1: **System Architecture of TSN in Wi-Fi Network [9]**

also connects to client nodes to have a parallel path to compare the time difference between access points and their assigned users, we do not discuss further PTP protocol further in this learning report.

b) Access Point: The configuration of access points utilizes OpenWifi [10] software-defined radio to provide controls for PHY and MAC layer, this enables the Time Synchronization Functions (TSF) counter in AP devices. TSF timer relies on the UTC times synchronized by PTP implementation, especially, when the PTP master is AP and the slave is the client, the consistent TSF timer allows more detectable offsets between wireless devices as the accurate synchronization from TSF as the reference point.

c) Client Node: The client nodes are Commercial off-the-shelf Wi-Fi card without additional configuration [10]. However, each node must support the synchronization function to receive the beacon-based packet to adjust the system time by comparing the transmission timestamps from AP with the received one, the logged timestamp stored in drivers to provide the elements to calculate the offsets in wireless devices. The beacon interval is important because the aggressive beacon

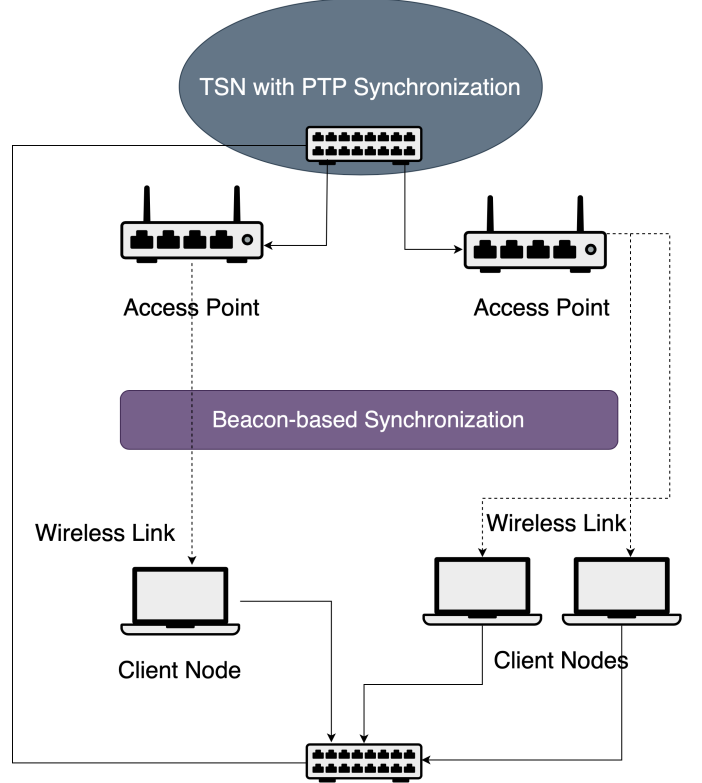


Fig. 2: **Measurement Setup of TSN in Wi-Fi Network [10]**

message period can lead to an overhead condition in the system [10], the author selected several beacon intervals to ensure the measurable results without causing a significant traffic load.

2) Evaluation Index: The reference experiment [10] selects three statistical indexes to evaluate the performance of the wireless network, the standard deviation σ_{ij} is the time differences in jitter between the client nodes, mean μ_{ij} is the mean time difference value and pr_{90} is the 90th of time difference from client nodes. The method of calculating the time difference is to subtract the received timestamp from each client.

3) Results: The original measurement contains 4 groups of data based on the evaluation index, however, we discuss the two results only where the beacon intervals are 100 ms and 500 ms for comparison purposes [10]. The standard deviation is 18.03 μs , the mean difference is 15.04 μs , and the 90th is

24 μs from the 100 ms case when the interval becomes 500 ms, σ_{ij} increases to 23.21 μ is 20.12 μs and pr_{90} is 35 μs .

The shorter interval can ensure more stable time differences based on the standard deviation and the 90th percentile from the measured data. However, the more important thing is that there is always a small offset even if time synchronization is complete among the client nodes, the reference [10] considers this time difference as the synchronization error in the wireless network of the experiment and the lower time difference value is, the more accurate the time synchronization performs in client nodes. They also intend to improve this accuracy by balancing the overhead in the setup from Figure 2 and they claim that 100 ms is the optimal choice for beacon interval.

4) *Limitations:* This case from reference [10] provides a practical evaluation of a TSN Wi-Fi network by utilizing a time synchronization technique. However, the case utilizes the statistical index to evaluate the time differences from the client nodes, not the end-to-end latency, it is nevertheless a valuable study material considering it indicates the stability of the TSN Wi-Fi network in a controlled environment.

Furthermore, this case intends to avoid overhead with beacon-based messages under an excessively low interval, which indicates that this setup has its bottleneck with beacon-based messages and inappropriate intervals can increase the latency in an overhead condition. The reference specifies that the optimization of overhead is for future research.

Finally, this case utilizes PTP Linux implementation but the core TSN Wi-Fi infrastructure as Figure 1 illustrates, therefore, it is not a comprehensive reference model for TSN in a Wi-Fi implementation. However, the value of this experiment is to implement time synchronization to obtain the measurement, which is related to the research orientation that we intend to concentrate on in the following stage.

III. TSN FOR 5G APPLICATIONS

This chapter discusses the second wireless application that TSN can implement which is the 5G cellular network. There are three sections covering this topic. The first scenario is an industrial application from CumuCore [11], where we introduce the fundamental elements of their solution. The second section illustrates the general system architecture when TSN is in a simplified 5G cellular network. The third section summarizes a simulation process from a reference document [12] as a case study.

A. Industrial Application in CumuCore

CumuCore is a Finnish company providing private mobile networking solutions for industrial purposes; it is operating a project related to TSN over 5G cellular networks to build deterministic wireless communication in an industrial environment. The latest 5G specification is compatible with a centralized TSN network model [12], where part of the 5G infrastructure works as a virtual TSN bridge (Figure 3).

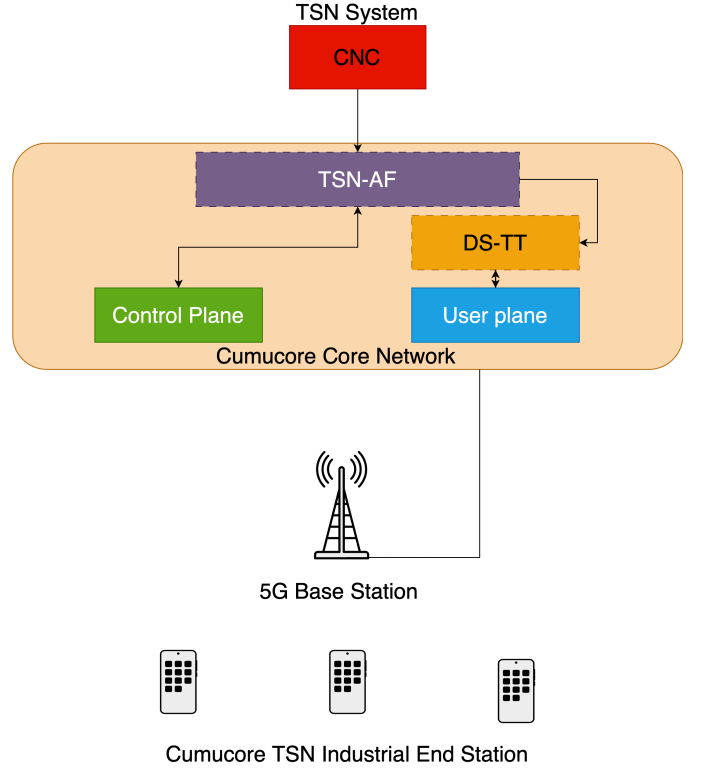


Fig. 3: System Architecture of TSN in 5G from CumuCore [11]

a) **CumuCore System Architecture:** The implementation of TSN in 5G from CumuCore is to integrate the CNC (Centralized Network Configuration) into the current 5G infrastructure for industrial users. CNC under CumuCore's structure contains the minimum and maximum delay in different cellular traffic categories and valid time in different network devices. To facilitate a clearer understanding, Figure 3 provides a simplified overview of the CumuCore implementation, we introduce each component within the TSN-AF, and DS-TT is introduced at a general level, without an in-depth technical exposition.

b) **TSN-AF:** TSN Application Function (TSN-AF) is an essential interface between the 5G system and CNC in TSN, which needs to translate the parameters from TSN to 5G core networks for seamless coordination in scheduling the traffic; there are specifications to define the TSN capabilities in terms of TSN-AF related to its functions in IEEE 802.1Q-2018 [11]. The TSN-AF translates the end station configuration by receiving parameters from the UE and DS-TT and forwarding them to the CNC, it uses a mapping table to classify traffic and assign corresponding priorities during the configuration process. CNC in this CumuCore infrastructure also calculates the deterministic paths and schedules in the 5G cellular network to ensure bounded latency and reliability. When the 5G system enables the Quality of Services (QoS), the scheduled traffic and filtering policies from CNC become the 5G QoS flows through the TSN-AF.

c) **DS-TT**: Device-Side TSN Translator (DS-TT) is responsible for providing timestamps and data frame functions to UE (User Equipment) within a 5G cellular network; it is essential to ensure that UE or TSN end stations are compatible with TSN functions. Cumucore details a time synchronization technique that extracts the Timing Synchronization information (TSi) from the Suffix field of generalized Precision Time Protocol (gPTP) messages [11], this approach allows precise time synchronization to be maintained across the network, which is crucial for guaranteeing deterministic communication and low latency in TSN-enabled networks. Furthermore, the case study analyzes the realistic performance and behavior of a TSN-enabled 5G network.

B. Case-Study

1) **Simulation Setup**: We outline this case study based on a reference work [12] where they conduct a simulation by configuring a 5G-TSN environment within a Network Simulator 3 (NS-3) [13]. Figure 4 illustrates the components and connections from the simulation configuration, they consider the TSN Endstation as a robot from the manufacturer, which connects into a 5G UE through several TSN bridges with wired channels, then this UE transmits the data into another UE with gNB (Base station) with the wireless channel, then another UE receives it and forwards it into other TSN GM by several TSN bridges. The reference work defined two classes for nodes (devices) with 5G and TSN.

a) **TSN Node**: The nodes under the TSN class are implementing Carrier Sense Multiple Access (CSMA) channel supporting layer 2 communication to meet the requirement of the simulation. Generalized Precision Time Protocol (gPTP) is the message model used to communicate during simulation runs, which incorporates time error caused by factors including clock inaccuracies and network delays into the time synchronization process, instead of relying solely on jitter and clock granularity calculated from the PHY (Physical) layer.

b) **5G Node**: The 5G node needs to interconnect the communication between TSN and 5G as a gateway, CSMA channel also installs into them. Moreover, the 5G node also installs the TSN Translator (TSN-AF) function in order that CNC can gather the parameters and calculate the scheduling traffic from the 5G network. The core 5G network within the nodes is LENA module [12] and it uses UDP/IP packets to transfer the packets, the central frequency band is 28 GHz, bandwidth is 400 MHz, and packet size is 44 Byte.

There are a total of 100 simulations with 100 s for each run, the synchronization interval with the master clock is 125 ms, and there are also frequency offset, drift rate, and other parameters, but we cover these topics only at a general level in this learning report, without going into detailed explanations or dedicated analysis.

2) **Results**: The evaluation of this simulation utilizes synchronization and syntonization errors for 5G ingress and egress devices to determine the accuracy of the 5G-TSN network. The former represents the time differences between devices related to the clock time, and the latter refers to the frequency of

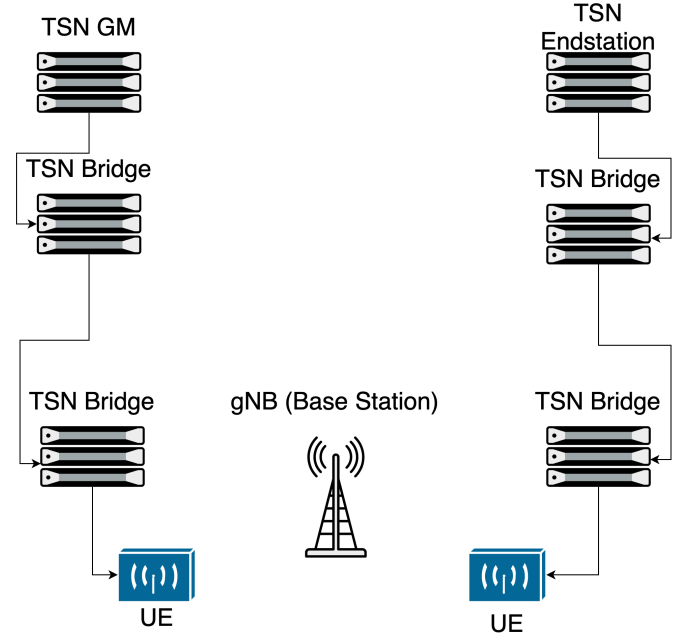


Fig. 4: Simulation Architecture from 5G-TSN Simulation [12]

the clock and manifests the stability for devices to maintain alignment with time.

a) **Synchronization Error**: The results show that the more TSN bridges that they add, the more accumulative synchronization error it produces. They explained that a small amount of TSN bridges do not produce a drastic variance and the primary component to have synchronization error is 5G internal core network [12]. The large number of TSN bridges generates more variance resulting in larger synchronization errors.

b) **Syntonization Error**: The frequency can drift or have offset in ingress and egress 5G device clocks. This error might affect the accuracy of clock synchronization in a 5G-TSN network, the maximum syntonization error that the reference claims is 750 ns [12]. One of the methods to mitigate large syntonization errors is to adjust the corresponding synchronization interval but not cause the system overload, the shorter synchronization interval can improve the syntonization error with less chance of having a large offset or drift from the device clocks.

Therefore, the simulation of the reference work intends to demonstrate the importance of appropriate configurations in a 5G system under a TSN implementation to serve industrial users, one of the potential solutions can be a private 5G network based on Cumucore service page [11], however, we do not introduce and elaborate private mobile network in this learning report.

3) **Limitations**: This case evaluates the performance of a 5G-TSN system with a simulation method, which can reflect the characteristics of this system, but it might not completely reproduce a realistic experiment, especially for wireless

channels because multiple factors can affect the final results. Additionally, this simulation also utilizes the synchronization and syntonization errors, not the end-to-end latency that we prefer to have. Finally, this reference work implements a comprehensive simulation with detailed configuration in NS-3, the results show a dependency between the synchronization error and 5G system parameters, not indicating if the 5G-TSN system meets the requirement of the industrial applications.

IV. TSN Wi-Fi AND 5G COMPARISON

This section presents a comparative analysis of two case studies on Time-Sensitive Networking (TSN) over Wi-Fi [10] and 5G [12], while previous sections describe their configurations and measurement methodologies, a direct comparison is necessary to highlight the key insights derived from both approaches. Therefore, this section is divided into two parts: the first compares the studies in terms of scope, feasibility, assumptions, and evaluation metrics, while the second discusses the observations and limitations of each study.

A. Comparison

a) Study Scope: The TSN for Wi-Fi study focuses on enhancing time synchronization using a low-overhead beacon mechanism in both intra-AP (within the same Access Point) and inter-AP (across multiple Access Points) scenarios, its primary objective is to improve synchronization accuracy at the mechanism level within IEEE 802.11 networks. In contrast, the TSN over 5G study investigates synchronization performance in an integrated TSN-5G architecture [12], emphasizing system-level behavior, it evaluates how synchronization and syntonization errors affect end-to-end performance in industrial communication scenarios.

b) Feasibility: The TSN for Wi-Fi study is validated through a physical testbed using real hardware and network traffic, which enhances its practical relevance. However, the evaluation is conducted in a controlled laboratory environment with limited network scale and minimal interference, while this setup demonstrates feasibility, it primarily reflects reproducibility under controlled conditions rather than large-scale or highly dynamic deployments.

In contrast, the TSN for 5G study is conducted with simulations using the Network Simulator 3 (ns-3) platform. Although this approach lacks validation through real-world implementation, it enables flexible configuration, scalability, and systematic analysis of key parameters that would be difficult to reproduce experimentally. However, the study does not provide evidence of practical feasibility, as no hardware-based implementation or experimental validation is presented, and we cannot conclude if this topology is reproducible based on their research article solely.

c) Critical Assumptions: The TSN for Wi-Fi study assumes that Access Points support low-level control of the Timing Synchronization Function (TSF) and allow the implementation of beacon follow-up mechanisms, this restricts applicability to software-defined or modified hardware platforms. Furthermore, inter-AP synchronization is assumed to

be provided via a wired network, implying dependence on an external time reference. The study also assumes that timestamps from client side at the software level is sufficiently accurate, despite potential processing delays. Additionally, the beacon follow-up mechanism is assumed to compensate for channel access delays introduced by the non-deterministic CSMA/CA protocol. Finally, the evaluation is conducted under controlled conditions with limited network scale and mobility, minimizing interference and variability.

On the other hand, the TSN for 5G study adopts several model-based assumptions. The 5G system is abstracted as a logical TSN bridge, simplifying the interaction between wired and wireless domains. Accurate timestamps at ingress and egress access points are assumed, even though these timestamps are generated by physically separate clocks. The clock behavior is modeled using simplified parameters, including constant frequency offset and drift, based on an ideal simulation time reference. Periodic synchronization is assumed to maintain system stability, while key network parameters including frequency band, bandwidth, numerology, and packet size are fixed to create a controlled simulation environment. These assumptions enable analytical evaluation but abstract away real-world aspects containing interference, mobility, and traffic variability.

d) Evaluation Metrics: The TSN for Wi-Fi study evaluates synchronization accuracy using microseconds (μs) to scale error measurements between client nodes. Statistical metrics including mean, standard deviation, and percentiles (25^{th} , 75^{th} , and 90^{th}) are used to characterize performance under different beacon intervals.

As for the TSN for 5G study focuses on synchronization and syntonization errors at the nanosecond (ns) scale, particularly at ingress and egress access points within the network. The evaluation emphasizes worst-case and end-to-end synchronization accuracy, considering parameters such as user equipment (UE) frequency offset, subcarrier spacing, and the number of TSN bridges. Unlike the Wi-Fi study, which focuses on statistical behavior, the 5G study targets deterministic performance requirements for industrial applications.

B. Observation and Limitations

a) TSN for Wi-Fi: The study demonstrates that beacon follow-up message mechanisms can significantly improve synchronization accuracy in both intra-AP and inter-AP scenarios [10]. The results indicate that synchronization performance is influenced by factors such as beacon interval and Access Point accuracy, while remaining relatively robust under network load conditions.

However, the applicability of this approach is constrained by several factors. The requirement for modified hardware and low-level TSF control limits deployment in standard Wi-Fi networks. Additionally, the evaluation is conducted in a controlled environment with limited interference, small network scale, and restricted mobility. Consequently, the performance under large-scale, highly dynamic, or interference-prone environments remains uncertain. Furthermore, due to the

inherent non-deterministic nature of IEEE 802.11, strict TSN guarantees cannot be fully ensured.

b) *TSN for 5G*: The study shows that even small frequency offsets in integrated 5G TSN networks can lead to significant synchronization errors, potentially challenging the strict requirements of industrial applications. The results highlight the sensitivity of synchronization performance to synchronization errors and internal 5G synchronization mechanisms.

Nevertheless, the study is based entirely on simulation using NS-3, which limits its ability to reflect real-world performance. Practical factors including wireless interference, mobility, and network congestion are not fully considered. Additionally, several parameters are assumed to be constant, which may not hold in realistic deployments. As a result, while the study provides valuable theoretical insights, further experimental validation is required to confirm its feasibility in real industrial environments.

V. SIONNA AND PHYSICAL LAYER ABSTRACTION

This section constitutes the second part of the learning report and presents a practical implementation of TSN-related research. The objective is not to develop a sophisticated wireless model from scratch, but rather to select and utilize a key tutorial from the Sionna instructional materials [5]. Accordingly, this section focuses on the study and analysis of an existing Physical Layer Abstraction framework provided by Sionna. The first subsection outlines the challenges encountered when installing the Sionna library on a MacOS platform. The second through fourth subsections describe the configuration, simulation, and evaluation processes associated with the Physical Layer Abstraction model in Sionna. Finally, the last subsection discusses potential future simulation models related to TSN.

A. Installation of Sionna and Potential Issues

We install Sionna libraries on Jupyter lab to simulate all the projects. The compulsory requirements for Sionna are Python and TensorFlow, however, it is important to notice that TensorFlow can only support certain versions of Python [14]. It supports Python 3.9 to 3.12 at the time of this learning report. We install TensorFlow with Python 3.10 to run Sionna successfully on an Intel Macbook Pro running MacOS 26.3.1. There are some challenges to overcome when we attempt to install Python 3.10 because TensorFlow will not work if the version of Python does not match its requirements.

The recommendation is to use **Conda** commands to create a Python environment with a specific version due to the requirements of TensorFlow, we provide the instructions with Python 3.10 because this is the version we can *pip install* TensorFlow without any system errors. Afterwards, we must install TensorFlow to have the foundations for Sionna modules, we use TensorFlow 2.13 in the instructions because this version does not trigger any issues for the simulations from to this learning report. Finally, we install Sionna and all three major modules including RT (Ray Tracing), PHY (Link

Level), and SYS (Physical Layer Abstraction) with *pip install* respectively. There is a list of instructions available:

```
conda create -n sionna-env python=3.10 -y
pip install tensorflow==2.13
pip install sionna
```

Once the process of installing Python 3.10, TensorFlow 2.13 and Sionna modules is complete, we must initiate a virtual environment with **Conda** to implement some wireless models in Sionna, we choose Jupyter Lab to have all the scripts and launch the simulation. The additional instructions for these processes are:

```
conda activate sionna-env
jupyter lab
```

Once Jupyter Lab or Notebook launches a tab in the local browser, creating a blank file, users can simulate wireless communication by running scripts based on Sionna libraries. We implement all the simulations and measurements in this learning report with **Sionna v1.2.1**.

B. Physical Layer Abstraction

There are two levels of wireless networking simulations, including link and system levels [15]. The first one aims to construct and configure comprehensive and realistic physical modules to facilitate signal transmission. The second one intends to have a large number of nodes, computed tables, and evaluation metrics to implement a networking scenario without executing the complex physical architecture, saving experimental time due to computational constraints. Therefore, this section demonstrates a system-level simulation based on the aforementioned definition although Sionna is completely capable of constructing physical layers to implement a simulation, this learning report concentrates on a wireless model with the category of system-level simulation.

Figure 5 illustrates the structure of a PHY abstraction in Sionna to construct a related wireless communication model. This technique intends to execute high-layer simulations, bypassing the physical signal processing while preserving accuracy in the evaluation of transportation quality. Hexagonal topology is an extensive network geometry for cellular implementation, where each cell has a base station to support user equipment in this cell. Sionna PHY channel contains multiple signal processing elements including forward error correction (FEC), multiple input multiple output (MIMO), and orthogonal frequency division multiplexing (OFDM), however, this learning report does not intend to explore the Sionna PHY channel comprehensively. Signal-to-interference-plus-noise is the value representing the effect by combining the transmission power and internal noise to provide an input for physical abstraction [16].

a) *Effective SINR*: codeword symbols can obtain various SINR values due to different subcarriers, streams, or time instances [15], leading to an inconsistent block error rate (BLER) that an AWGN channel produces. The effective SINR

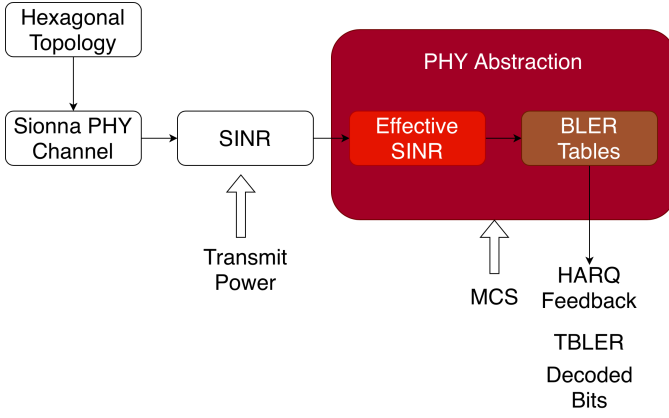


Fig. 5: Sionna PHY Abstraction Structure [16]

is the special one that generates the same BLER among them, an effective SINR for a single user with the configured $SINR_i$ in equation (1):

$$-\beta \log \left(\frac{1}{N} \sum_{i=1}^N e^{-\frac{SINR_i}{\beta}} \right), \quad (1)$$

where β is a scaling parameter whose value depends on the modulation and coding scheme (MCS) applied to the simulation. In this learning report, we discuss all the configurations and evaluations based on the selected MCS tables from Sionna tutorial [16], therefore, β is a calibrated value in equation (1), N represents the number of time or frequency resources, and $SINR_i$ is the signal-to-interference-plus-noise ratio from each resource unit.

b) BLER Table: A pre-computed Block Error Rate (BLER) Table is a two-dimensional interpolated grid consisting of SINR and code block size (CBS) values, which provides a reference due to the Modulation Code Scheme (MCS) can result in unique SINR and BLER in an additive white Gaussian noise channel (AWGN) [16]. The simulation can predict the error rate of a certain codeword by selecting the corresponding BLER, meanwhile, setting the threshold of BLER (e.g., 10%) can filter the transportation packages that are unreliable due to their BLER.

In a word, PHY abstraction is a process of computation by utilizing effective SINR (equation (1)) and obtaining BLER tables by selecting optimal MSC values. There are three output after the BLER tables section including Hybrid Automatic Repeat Request (HARQ) feedback to indicate if the transport block reaches the receiver, Transport BLER (TBLER) represents the possibility that at least one incorrect decoded code block is in a transport process.

C. Simulation Configurations

This section describes the procedures to implement a physical layer abstraction on a Jupyter Lab based on the tutorial [16]. We summarize and emphasize several critical parts instead of listing the entire script from the official tutorial.

The initial step is to import all the required and relevant external libraries. There is a library named **PHYAbstraction** from the Sionna system module, we can instantiate a PHYAbstraction object relying on this library. Then the category defines the uplink (0) and downlink (1), MCS table and MCS indexes are the combinatorial parameters for 5G NR [4], MCS table 1 represents the 64 Quadrature Amplitude Modulation (QAM), and the MCS index is the code rate of the selected MCS table. Therefore, the MCS index 15 refers to the 15th row of its table, where the target code rate is 616×1024 , and the spectral efficiency is 2.4063 [16]. The scripts plot two figures with MCS 15 and MCS 17 from the MCS table 1 to emphasize that MCS is not the only factor to affect BLER, the longer code block size can result in lower BLER because of its robustness for error bits.

The second step is to obtain BLER from a two-dimensional grid consisting of SINR and code block size (CBS) [16]. This process returns the transportation block error rate (TBLER) that has at least one incorrectly decoded code block, the tutorial selects the SNR 9 dB under MCS index 15 with code block size 80 to obtain BLER value 0.0511, which is the precomputed result within the Sionna System library. By giving specific parameters, it is worth noticing that generating a new BLER is feasible in Sionna by inputting a new code block size and SINR to compute a new BLER table.

The last step is the configuration of the number of decoded bits, Hybrid Automatic Repeat Request (HARQ), before computing the effective SINR and Transport BLER. The tutorial set 80 users with 14 OFDM symbols, 1024 subcarriers, and only one stream for each User Equipment (UE). Moreover, the target BLER is 10% because we want to find the best MCS index for each user that can retain the Transport BLER within the reliability through the Inner Loop Link Adaptation (ILLA) module [16]. We can generate each stream from all 80 users to launch the simulation to obtain the number of decoded bits, HARQ feedback, and TBLER as Figure 5 shows.

D. Evaluation and Observations

We have a mathematical definition of Effective SINR in equation (1), in this section, we collect all the computed effective SINR values from all 80 users and visualize them with the arithmetic mean and effective SINR to observe the condition of channels per user.

Figure 6 illustrates the results of the simulation of a PHYAbstraction. The horizontal axis represents 80 users in descending order from the best to worst SINR in every stream with all OFDM subcarriers, the vertical axis represents the SINR value from low to high, the calculated effective SINR is the black solid line, the arithmetic mean is the red dashed line, and the yellow scatters are the individual SINR samples from each user.

We can observe that the arithmetic mean is almost identical to effective SINR from the first 30 users with less than 10 dB, after this point, effective SINR remains rising but the increasing rate is significant slower than arithmetic mean because the exponential effect based on equation 1, the large

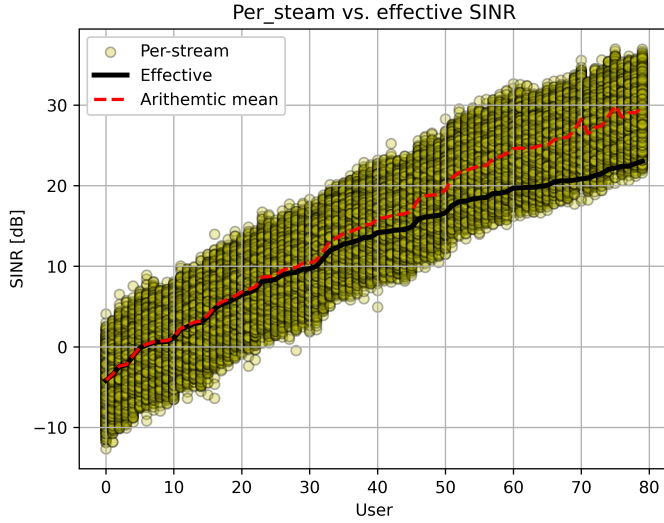


Fig. 6: Measurement Plot Sorted by SINR Values Per User [16]

$SINR_i$ can lead the part where $e^{-\frac{SINR_i}{\beta}}$ becomes greater than 0 but less than 1, resulting in effective SINR reduces its growing rate. In contrast, the users who has lower SINR has less impact on the difference between arithmetic mean and effective SINR due to the effect of small $SINR_i$ can be trivial on $e^{-\frac{SINR_i}{\beta}}$ that is near value 1, therefore, the output value can remain close to arithmetic mean.

The tutorial defines that a good channel reflects effective SINR shifts to the lower SINR scatterers among the users [16]. That indicates that the effective SINR can still rise even with low SINR value because this parameter can visualize that the limitation of a channel can be dependent on the weaker subcarriers, but arithmetic cannot be the indicator since it does not indicate this condition with users with high SINR. It is important to note that figure 6 is our simulation result, unlike the tutorial plot, our measurements shows an early stage where effective SINR stays below the arithmetic mean where the tutorial has a separate point for two metrics from the 50th users at around 22 dB. We speculate that the generated SINR scatters are not the same replica samples as the tutorial, in other word, the channel condition between our simulation and tutorial has minor differences, but the trending of effective SINR and arithmetic mean is identical. Finally, the selected MCS table and MCS indexes can also determine the measurement plot, but we will explore that in future implementations.

VI. CONCLUSIONS

A. Conclusions

This learning report investigates TSN implementation across two wireless domains, namely IEEE 802.11 and 5G cellular networks. It presents the fundamental concepts of TSN, IEEE 802.11, and 5G, supplemented by relevant industrial context, system architecture illustrations, and functional descriptions of key components. Evaluations from prior studies [10], [12] are incorporated to support the analysis of TSN-enabled wireless

scenarios. The results indicate that, in TSN-enabled Wi-Fi systems, shorter beacon intervals reduce time synchronization discrepancies among devices, while in 5G-TSN scenarios, minor parameter offsets can lead to significant variations in synchronization error. Furthermore, limitations in the referenced studies are identified, including the absence of end-to-end latency as a performance metric and the lack of comprehensive modeling of real-world wireless interference. In addition, a system-level simulation is conducted using Sionna to analyze a physical layer abstraction process. By following the tutorial configuration, a comparable result is reproduced, illustrating the trend between effective SINR and arithmetic mean for channel condition evaluation across 80 users per stream with 1024 OFDM subcarriers. The findings show that the effective SINR begins to fall below the arithmetic mean beyond approximately the 30th user and above the 10 dB threshold, indicating that, under favorable channel conditions, the effective SINR is strongly influenced by weaker subcarriers.

B. Future Research

For future research, Sionna [5] will be utilized to simulate more complex wireless environments in which TSN is integrated with either IEEE 802.11 (Wi-Fi) or 5G cellular networks. In addition, laboratory-based experiments may be conducted to measure end-to-end latency. This learning report does not cover specific Wi-Fi standards such as Wi-Fi 5, 6, or 7, as the availability of simulation tools and hardware platforms within the department must first be ensured. Similarly, the accessibility of cellular infrastructure must be evaluated prior to conducting TSN-related experiments in 5G networks.

A networking framework known as Reliable Available Wireless (RAW) [17] is currently under development, building upon the principles of Deterministic Networking (DetNet). As a next step, this work intends to develop a wireless model based on the RAW draft specifications, with the objective of contributing to ongoing research in this field.

It should be noted that the original Physical Layer Abstraction tutorial was unavailable at the time this learning report was prepared. An updated version is accessible https://github.com/NVlabs/sionna/blob/v2.0.0/tutorials/sys/PHY_Abstraction.ipynb, however, it introduces modifications to the implementation. Due to these differences, this report does not include simulation script, as inconsistencies between versions may result in errors.

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